

Preserving Family Album Photos with the HoloLens 2

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ABSTRACT

Following the leap set by the introduction of digital photography and spread of social media, the custom of creating and nourishing a family photo album is steadily declining. The feeling of waving goodbye to an old technology may hence explain why collectors and museums are increasingly searching for such type of photographs. In such scenario, a barrier set to the preservation and analysis of the enormous source of information is the unavailability of a fine-grained digitization and cataloguing infrastructure. An opportunity to resolve such issue is offered by augmented reality (AR): we envision an AR-application capable of individuating and segmenting photos, while browsing through a family album. In addition to this digitization step, we also include a second one where the digitized images are analyzed, tagging them with meta-data pertaining their presumed socio-historical context and their date. The addition of such meta-data is key not only from a cataloguing point of view, but also from a conservation perspective. In fact, analog photos are more often preserved when some information pertaining their subject and their date is known. To this aim, we experiment the use of the HoloLens 2 along with artificial intelligence paradigms.

Index Terms: Human-centered computing—Augmented Reality—Deep Learning—HoloLens 2

1 INTRODUCTION

The interest around analog family albums has recently gained momentum [7]. In fact, on one side the practice of collecting printed photos in family albums is declining. This decline, however, has triggered an increasing attraction from collectors and museums. Nevertheless, we may observe a general lack of supporting infrastructures, available to end users, for the digitization and cataloguing of the huge corpus of family album photographs. Augmented reality (AR) may respond such need. As a proof, we designed a HoloLens 2 application that: (a) digitizes the photos watched by a user in a family album, and, (b) analyzes the digitized images tagging them with meta-data representing their presumed socio-historical context and date. The addition of such meta-data is key not only from a cataloguing perspective, but also from a conservation point of view, as analog photos are more likely to be conserved when the scene and the people they portrait, as well as their date, is known [7].

This contribution follows a stream of works that have proposed the association of metadata to visual content in AR. Authors in [6], for example, described how AR-based computing systems could be put to good use to associate additional layers of knowledge to real cultural experiences. The HyperReality AR application, has been proposed to weave existing environments to digital content to enhance everyday activities [1, 3]. Such works paved the way to how, presumably, the WebXR will be put to good use in the near

future, exhibiting phydigital spaces where digital content is strictly intertwined to physical one [5].

We here describe a HoloLens 2 prototype application which mixes AR and Computer Vision paradigms for the benefit of family album preservation [9]. A visual concept of such application is reported in Figure 1. The remainder of this paper describes the main components of the application and its functionalities.

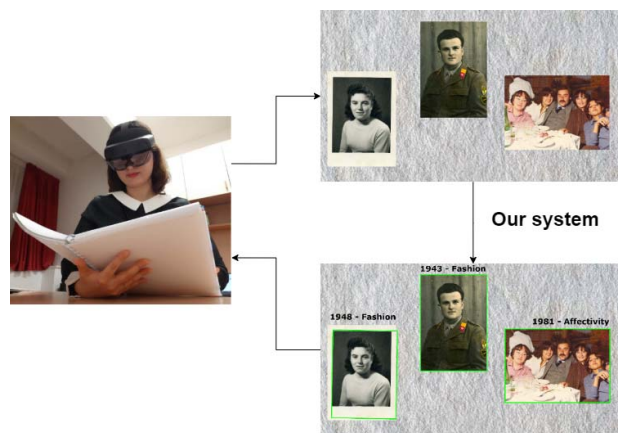


Figure 1: Application concept

2 SYSTEM ARCHITECTURE

The image classification and organization system will be based on the use of different tools: (a) an augmented and virtual reality device, namely HoloLens2, used to track a user's view and to visualize photo meta-data, (b) an image processing library to segment and process the pictures appearing in the scene, and, (c) two deep learning models to classify the segmented images. The general application is based on a client/server architecture, organized as follows.

The **client application** runs on the HoloLens2 device. The main aims of the client side are to capture the user view and visualize the information returned by the server about the date and the socio-historical context classification of each image in the view.

The **server application** infers the best classes resorting to deep learning algorithms, taking as input the user view sent by the client. To make deep learning models work at best, a preliminary step of image isolation is required to feed them with the target images only. The inferred information are then sent back to the client to be visualized on its interface in AR.

The HoloLens2, exploiting its locatable camera (RGB color camera), can access and control the real-time view of a user employing the Windows Media Capture and Media Foundation APIs. The usage of an image processing library is required in order to individuate, extract and process the region of interest in the user's view. The image processing library that has been chosen for this project is one of the most popular ones: OpenCV [4].

On the client side of this system, video frames (as images) are captured from the camera live stream of the device and sent to the

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server. At this point the server isolates target images in the user view via OpenCV image processing algorithms, as described in Section 2.1, and feeds them to deep learning models. The deep learning models, in turn, produce their classification. Finally, the server returns the classification results to the client side to be visualized on the HoloLens 2 interface.

2.1 Image pre-processing

Image pre-processing is a fundamental part of our project and it is applied to each frame extracted by the HoloLens2 locatable camera. The main aim of this step is to detect, extract and improve image targets before sending them to classification models. The first image pre-processing step consists in applying several transformation on the same frame in order to make the image target detection to be done properly: (a) a noisy removal and edge-preserved filter (e.g. bilateral filtering), (b) resizing and (c) conversion from RGB to grayscale space.

The second process includes detecting image target contours exploiting an automatic canny edge detector whose results are further improved with a closing operation to correct gaps between detected edges [2]. At this point the polygons, whose sides are defined by the detected contours, are approximated. In order to avoid bad polygons derived from noises or shadows in the scene, custom filters have been designed basing on the area and the number of sides. For instance, only those polygons which can be approximated with 4 sides (i.e. rectangles) are assumed to be target images.

Despite we expect that the target images are placed on a planar surface (e.g. photo albums and walls), a user is free to move around the scene watching them from different perspectives. Therefore, these extracted images need to be adjusted before employing deep learning models. To do so, a perspective warping transformation is applied to each target image. Finally the warped images are resized and evaluated by deep learning models. The warping algorithm returns the same image even when the user is watching the same picture from different points of view. This means that the entire process can be optimized guessing only once for all the classification related to a certain picture (e.g., a hash map is constructed where the key is the hash of the image and the value is its classifications).

2.2 Infer the date and the socio-historical context with deep learning models

As previously stated, once the images are available in a digital form, two deep learning models called, respectively, IMAGO DATING and IMAGO SOCIO HISTORICAL CLASSIFIER, developed by the authors of [8], infer their most probable date and socio-historical context. These models were trained specifically on analog photo coming from family albums and are able to classify both the date and the socio-historical context of a digitized analog pictures with an accuracy that falls slightly below the 70% threshold. In particular, the year range supported by the IMAGO DATING classifier is the [1930,1999] time frame, while the socio-historical classes available through the IMAGO SOCIO HISTORICAL CLASSIFIER amount to work, free-time, motorization, music, fashion, affectivity, rites, school, politics. Once these two information are individuated, the server returns to the client a vector containing $n \times 2$ data, where n is the number of previously detected pictures.

2.3 User supported actions

A user is asked to choose and attach the analog images of interest to a planar surface, based on some general rules: (a) the background of the surface should be simple, (b) the photographs should be in rectangular shape, but they can vary in size, (c) the arrangement of the photographs, regarding their position and orientation on the surface, can be random as long as the pictures do not overlap, and, (d) all images should be observable in a single field of view of the device.

Respecting these rules, a given user can wear the HoloLens2 device and start looking at the pictures. After few seconds, the user can visualize the detected bounding boxes along with the classifications of each target image, as depicted in Figure 1. As stated in Section 2 the user will see the classifications related to the date and the socio-historical context in which the photo was taken. Moreover, the system allows the user to interact with the visualized content. For example, the user has the ability of attaching extra information to each particular image including a voice comment, suggesting a different date and/or socio-historical context. These interaction experiences have been designed adopting interaction models that are common in mixed reality apps that combine hand, eye-gaze and natural language to provide a multi-modal interaction experience that may potentially provide instinctual experiences for the user.

3 CONCLUSIONS AND FUTURE WORKS

We here presented a new way to digitize and catalogue old analog pictures coming from family albums with the adoption of state of the art technologies. Firstly, we motivated our work considering the research problems today present in the community of photography studies, where the recovery analog family photo albums has become a compelling line of work.

In particular, we designed and implemented approach based on the use of an AR device, namely the HoloLens 2, and Computer Vision/Deep learning algorithms.

Future developments of our prototype will include the use of eye-tracking paradigms, as those offered by the HoloLens 2, to individuate in real time the focus of attention of a user. With an analysis of such data we will be able to acquire information regarding the user's interests and, in turn, extend our application to provide further information in relation to such interests.

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